

Degree: B.Tech. Semester: III Course work

END-SEMESTER EXAMINATION (Odd), Nov-Dec 2025

Course Title: Probability Theory and Random Process

Course Code: ECECC303/VTECC303/ECECC06/EIECC06

Max. Marks: 50

Time: 3 Hours

Note: - Attempt all five questions. Missing data/information, if any, maybe suitably assumed & mentioned in the answer.

Q.No.	Question	Marks	CO
Q1	Attempt any 2 parts of the following:		
1a	A signal may be sent through two parallel paths P_1 and P_2 . P_1 : used 70% of the time, success rate = 0.9 P_2 : used 30% of the time, success rate = 0. If a signal is successfully received, find the probability that it was sent through P_1 . Also compute the total probability of failure.	5	CO1
1b	A continuous random variable X has a probability density function (PDF) given by $f(x) = \begin{cases} kx^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ Find (i) the value of k , (ii) $P(X > 1)$, (iii) the cumulative density function (CDF) of X .	5	CO1
1c	One of two Poisson models is chosen randomly for count X : Poisson with $(\lambda = 1)$ is chosen with a probability 0.4, and Poisson with $(\lambda = 4)$ is chosen with a probability 0.6. i. Find $P(X = 0)$. ii. Given that $X = 0$, what is the probability that the chosen model was Poisson with $(\lambda = 1)$?	5	CO1
Q2	Attempt any 2 parts of the following:		
2a	Joint probability mass function (PMF) of $X \in \{0,1,2\}$ and $Y \in \{0,1\}$ is $P_{X,Y}(x,y) = C(x+2y)$ for given values of X and Y , and zero elsewhere. i. Find the constant C . ii. Find marginal PMFs, $p_X(x)$ and $p_Y(y)$. iii. Are X and Y independent?	5	CO2
2b	Let X have PDF $f_X(x) = 2x, 0 < x < 1$. i. Find the MGF $M_X(t)$. ii. Using $M_X(t)$, compute $E[X]$ and $\text{Var}(X)$. iii. If $Y = 3X + 2$, find $M_Y(t)$.	5	CO2
2c	A digital communication system transmits binary signal $S \in \{+1, -1\}$ which are equiprobable. Received signal $R = S + N$, where $N \sim \mathcal{N}(0, \sigma^2)$. i. Find $f_{RS}(r s)$ and ii. Find $P(S = +1 R = r)$.	5	CO2

Q3	Attempt any 2 parts of the following:		
3a	Given random variable X and Y . Find the PDF and CDF of i. $S = \max(X, Y)$ and ii. $T = \min(X, Y)$.	5	CO3
3b	State Markov and Chebyshev inequalities. Provide a step-by-step derivation of the Chebyshev inequality.	5	CO3
3c	Define Central Limit Theorem and explain its significance through a real life example.	5	CO3
Q4	Attempt any 2 parts of the following:		
4a	Given a random process consisting of two sequences of independent random variables defined by $Z_{2k+1} = X_k$ and $Z_{2k} = Y_k$, where $X_k = \begin{cases} 1, & \text{with probability } 0.5 \\ -1, & \text{with probability } 0.5 \end{cases}$ $Y_k = \begin{cases} 1, & \text{with probability } 0.8 \\ -2, & \text{with probability } 0.2 \end{cases}$ Check whether the process $\{Z_k\}$ is Stationary or Wide-sense stationary (WSS). Show all steps clearly for proper reasoning.	5	CO4
4b	$X(t) = A \cos(2\pi t)$, where A is a Gaussian random variable with mean '0' and variance σ^2 . Find the mean, autocorrelation, and autocovariance of $X(t)$.	5	CO4
4c	What do you understand by an i.i.d. random process and sum process? Define the Binomial Counting Process S_n and hence find the expression for probability $P(S_{n_1} = y_1, S_{n_2} = y_2)$.	5	CO4
Q5	Attempt any 2 parts of the following:		
5a	The auto correlation functions of two signals are given below: i. $R_X(\tau) = e^{-4 \tau }$ ii. $R_Y(\tau) = 2 \cos(\pi\tau)$ For each case, compute the Power Spectral Density (PSD) and comment on the total power of these signals.	5	CO5
5b	Let $Y(t) = X(t) + X(t-b)$. If $Y(t)$ is passed through an LTI system with impulse response $h(t) = e^{-at}u(t)$, and $Z(t)$ is the output of the LTI system, find the PSD of Z in terms of PSD of X .	5	CO5
5c	Let $Y_k = X_k + N_k$, where X_k and N_k are independent random variables representing transmitted signal and noise signal, respectively, where $E[X_k] = 0, E[N_k] = 0, E(X_k^2) = \sigma_X^2, E(N_k^2) = \sigma_N^2$. Comment on the stationarity of Y_k . Also, compute the PSD of Y_k .	5	CO5